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Methods and devices for utilizing bondable materials

Abstract

The invention primarily relates to fastening and stabilizing tissues, implants, and/or bondable materials, such as the fastening of a tissue and/or implant to a bondable material, the fastening of an implant to tissue, and/or the fastening of an implant to another implant. This may involve using an energy source to bond and/or mechanically to stabilize a tissue, an implant, a bondable material, and/or other biocompatible material. The invention may also relate to the use of an energy source to remove and/or install an implant and/or bondable material or to facilitate solidification and/or polymerization of bondable material.

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5423796	12/1994	Skikhman	N/A	N/A
5431670	12/1994	Holmes	N/A	N/A
5439470	12/1994	Li	N/A	N/A
5441538	12/1994	Bonutti	N/A	N/A
5443512	12/1994	Parr	N/A	N/A
5447503	12/1994	Miller	N/A	N/A
5449372	12/1994	Schmaltz	N/A	N/A
5449382	12/1994	Dayton	N/A	N/A
5451235	12/1994	Lock	N/A	N/A
5453090	12/1994	Martinez	N/A	N/A
5456722	12/1994	McLeod	N/A	N/A
5458653	12/1994	Davison	N/A	N/A
5462561	12/1994	Voda	N/A	N/A
5464424	12/1994	O'Donnell	N/A	N/A
5464426	12/1994	Bonutti	N/A	N/A
5464427	12/1994	Curtis	N/A	N/A
5470337	12/1994	Moss	N/A	N/A
5472444	12/1994	Huebner	N/A	N/A
5474554	12/1994	Ku	N/A	N/A
5478351	12/1994	Meade	N/A	N/A
5478353	12/1994	Yoon	N/A	N/A
5480403	12/1995	Lee	N/A	N/A
5486197	12/1995	Le	N/A	N/A
5487844	12/1995	Fujita	N/A	N/A
5488958	12/1995	Topel	N/A	N/A
5496292	12/1995	Burnham	N/A	N/A
5496335	12/1995	Thomason	N/A	N/A
5496348	12/1995	Bonutti	N/A	N/A
5499382	12/1995	Nusinov et al.	N/A	N/A
5500000	12/1995	Feagin	N/A	N/A
5501700	12/1995	Hirata	N/A	N/A
5504977	12/1995	Weppner	N/A	N/A
5505735	12/1995	Li	N/A	N/A

5507754	12/1995	Green	N/A	N/A
5522844	12/1995	Johnson	N/A	N/A
5522845	12/1995	Wenstrom	N/A	N/A
5522846	12/1995	Bonutti	N/A	N/A
5527341	12/1995	Gogolewski	N/A	N/A
5527342	12/1995	Pietrzak	N/A	N/A
5527343	12/1995	Bonutti	N/A	N/A
5529075	12/1995	Clark	N/A	N/A
5531759	12/1995	Kensey	N/A	N/A
5534012	12/1995	Bonutti	N/A	N/A
5534028	12/1995	Bao	N/A	N/A
5540718	12/1995	Bartlett	N/A	N/A
5542423	12/1995	Bonutti	N/A	N/A
5545178	12/1995	Kensey	N/A	N/A
5545180	12/1995	Le	N/A	N/A
5545206	12/1995	Carson	N/A	N/A
5549630	12/1995	Bonutti	N/A	N/A
5549631	12/1995	Bonutti	N/A	N/A
5556402	12/1995	Xu	N/A	N/A
5569252	12/1995	Justin	N/A	N/A
5569305	12/1995	Bonutti	N/A	N/A
5569306	12/1995	Thal	N/A	N/A
5571190	12/1995	Ulrich et al.	N/A	N/A
5573517	12/1995	Bonutti	N/A	N/A
5573538	12/1995	Laboureau	N/A	N/A
5573542	12/1995	Stevens	N/A	N/A
5575801	12/1995	Habermeyer	N/A	N/A
5580344	12/1995	Hasson	N/A	N/A
5584835	12/1995	Greenfield	N/A	N/A
5584860	12/1995	Goble	N/A	N/A
5584862	12/1995	Bonutti	N/A	N/A
5591206	12/1996	Moufarrege	N/A	N/A
5593422	12/1996	Muijs Van De Moer	N/A	N/A
5593425	12/1996	Bonutti	606/228	B29C 66/8322
5593625	12/1996	Rieble	N/A	N/A
5601557	12/1996	Hayhurst	N/A	N/A
5601558	12/1996	Torrie	N/A	N/A
5601595	12/1996	Schwartz	N/A	N/A
5607427	12/1996	Tschakaloff	N/A	N/A
5609595	12/1996	Pennig	N/A	N/A
5618314	12/1996	Harwin	N/A	N/A
5620461	12/1996	Muijs Van De Moer	N/A	N/A
5626612	12/1996	Bartlett	N/A	N/A
5626614	12/1996	Hart	N/A	N/A
5626718	12/1996	Philippe	N/A	N/A
5628751	12/1996	Sander	N/A	N/A
5630824	12/1996	Hart	N/A	N/A
5634926	12/1996	Jobe	N/A	N/A
5643274	12/1996	Sander	N/A	N/A

5643293	12/1996	Kogasaka	N/A	N/A
5643295	12/1996	Yoon	N/A	N/A
5643321	12/1996	McDevitt	N/A	N/A
5645553	12/1996	Kolesa	N/A	N/A
5645597	12/1996	Krapiva	N/A	N/A
5645599	12/1996	Samani	N/A	N/A
5649955	12/1996	Hashimoto	N/A	N/A
5649963	12/1996	McDevitt	N/A	N/A
5651377	12/1996	O'Donnell	N/A	N/A
5658313	12/1996	Thal	N/A	N/A
5658408	12/1996	Frantz	N/A	N/A
5660225	12/1996	Saffran	N/A	N/A
5662658	12/1996	Wenstrom	N/A	N/A
5665089	12/1996	Dall	N/A	N/A
5665109	12/1996	Yoon	N/A	N/A
5667513	12/1996	Torrie	N/A	N/A
5669917	12/1996	Sauer	N/A	N/A
5674240	12/1996	Bonutti	N/A	N/A
5681310	12/1996	Yuan	N/A	N/A
5681333	12/1996	Burkhart	N/A	N/A
5681351	12/1996	Jamiolkowski	N/A	N/A
5681352	12/1996	Clancy	N/A	N/A
5685820	12/1996	Riek	N/A	N/A
5688283	12/1996	Knapp	N/A	N/A
5690654	12/1996	Ovil	N/A	N/A
5690655	12/1996	Hart	N/A	N/A
5690676	12/1996	DiPoto	N/A	N/A
5693055	12/1996	Zahiri	N/A	N/A
5697950	12/1996	Fucci	N/A	N/A
5702397	12/1996	Gonie	N/A	N/A
5702462	12/1996	Oberlander	N/A	N/A
5707395	12/1997	Li	N/A	N/A
5713903	12/1997	Sander	N/A	N/A
5713921	12/1997	Bonutti	N/A	N/A
5718717	12/1997	Bonutti	N/A	N/A
5720747	12/1997	Burke	N/A	N/A
5720753	12/1997	Sander	N/A	N/A
5725541	12/1997	Anspach	N/A	N/A
5725556	12/1997	Moser	N/A	N/A
5725582	12/1997	Bevan	N/A	N/A
5730747	12/1997	Ek	N/A	N/A
5733306	12/1997	Bonutti	N/A	N/A
5735875	12/1997	Bonutti	N/A	N/A
5735877	12/1997	Pagedas	N/A	N/A
5735899	12/1997	Schwartz	N/A	N/A
5741282	12/1997	Anspach	N/A	N/A
5752952	12/1997	Adamson	N/A	N/A
5752974	12/1997	Rhee	N/A	N/A
5755809	12/1997	Cohen	N/A	N/A
5762458	12/1997	Wang	N/A	N/A

5766221	12/1997	Benderev	N/A	N/A
5769894	12/1997	Ferragamo	N/A	N/A
5772672	12/1997	Toy	N/A	N/A
5776151	12/1997	Chan	N/A	N/A
5779706	12/1997	Tschakaloff	N/A	N/A
5782862	12/1997	Bonutti	N/A	N/A
5785713	12/1997	Jobe	N/A	N/A
5792096	12/1997	Rentmeester	N/A	N/A
5797931	12/1997	Bito	N/A	N/A
5800537	12/1997	Bell	N/A	N/A
5807403	12/1997	Beyar	N/A	N/A
5810849	12/1997	Slavros	N/A	N/A
5810853	12/1997	Yoon	N/A	N/A
5810884	12/1997	Kim	N/A	N/A
5814072	12/1997	Bonutti	N/A	N/A
5814073	12/1997	Bonutti	N/A	N/A
5817107	12/1997	Schaller	N/A	N/A
5823994	12/1997	Sharkey	N/A	N/A
5824009	12/1997	Fukuda	N/A	N/A
5830125	12/1997	Scribner	N/A	N/A
5836897	12/1997	Sakural	N/A	N/A
5839899	12/1997	Robinson	N/A	N/A
5843178	12/1997	Vanney	N/A	N/A
5845645	12/1997	Bonutti	N/A	N/A
5851185	12/1997	Berns	N/A	N/A
5865834	12/1998	McGuire	N/A	N/A
5866634	12/1998	Tokushige	N/A	N/A
5868749	12/1998	Reed	N/A	N/A
5874235	12/1998	Chan	N/A	N/A
5879372	12/1998	Bartlett	N/A	N/A
5891166	12/1998	Schervinsky	N/A	N/A
5891168	12/1998	Thal	N/A	N/A
5893880	12/1998	Egan	N/A	N/A
5897574	12/1998	Bonutti	N/A	N/A
5899911	12/1998	Carter	N/A	N/A
5899921	12/1998	Caspari	N/A	N/A
5906579	12/1998	Vander Salm	N/A	N/A
5906625	12/1998	Bito	N/A	N/A
5908429	12/1998	Yoon	N/A	N/A
5911721	12/1998	Nicholson	N/A	N/A
5918604	12/1998	Whelan	N/A	N/A
5919193	12/1998	Slavitt	N/A	N/A
5919194	12/1998	Anderson	N/A	N/A
5919208	12/1998	Valenti	N/A	N/A
5919215	12/1998	Wiklund	N/A	N/A
5921986	12/1998	BOnutti	N/A	N/A
5925064	12/1998	Meyers	N/A	N/A
5928244	12/1998	Tovey	N/A	N/A
5928267	12/1998	Bonutti	N/A	N/A
5931838	12/1998	Vito	N/A	N/A

5931869	12/1998	Boucher	N/A	N/A
5940942	12/1998	Fong	N/A	N/A
5941900	12/1998	Bonutti	N/A	N/A
5941901	12/1998	Egan	N/A	N/A
5945002	12/1998	Leukes et al.	N/A	N/A
5947982	12/1998	Duran	N/A	N/A
5948000	12/1998	Larsen	N/A	N/A
5948001	12/1998	Larsen	N/A	N/A
5948002	12/1998	Bonutti	N/A	N/A
5951590	12/1998	Goldfarb	N/A	N/A
5957953	12/1998	DiPoto	N/A	N/A
5961499	12/1998	Bonutti	N/A	N/A
5961521	12/1998	Roger	N/A	N/A
5961554	12/1998	Janson	N/A	N/A
5964765	12/1998	Fenton	N/A	N/A
5964769	12/1998	Wagner	N/A	N/A
5968046	12/1998	Castleman	N/A	N/A
5968047	12/1998	Reed	N/A	N/A
5980520	12/1998	Vancaillie	N/A	N/A
5980559	12/1998	Bonutti	N/A	N/A
5984929	12/1998	Bashiri	N/A	N/A
5989282	12/1998	Bonutti	N/A	N/A
5993458	12/1998	Vaitekunas	N/A	N/A
5993477	12/1998	Vaitekunas	N/A	N/A
6007567	12/1998	Bonutti	N/A	N/A
6007580	12/1998	Lehto	N/A	N/A
6010525	12/1999	Bonutti	N/A	N/A
6010526	12/1999	Sandstrom	N/A	N/A
6017321	12/1999	Boone	N/A	N/A
6033429	12/1999	Magovern	N/A	N/A
6033430	12/1999	Bonutti	N/A	N/A
6045551	12/1999	Bonutti	N/A	N/A
6050998	12/1999	Fletcher	N/A	N/A
6056751	12/1999	Fenton, Jr.	606/232	A61F 2/0811
6056772	12/1999	Bonutti	N/A	N/A
6056773	12/1999	Bonutti	N/A	N/A
6059797	12/1999	Mears	N/A	N/A
6059817	12/1999	Bonutti	N/A	N/A
6059827	12/1999	Fenton	N/A	N/A
6063095	12/1999	Wang	N/A	N/A
6066151	12/1999	Miyawaki	N/A	N/A
6066160	12/1999	Colvin	N/A	N/A
6066166	12/1999	Bischoff	N/A	N/A
6068637	12/1999	Popov	N/A	N/A
6077277	12/1999	Mollenauer	N/A	N/A
6077292	12/1999	Bonutti	N/A	N/A
6080161	12/1999	Eaves, III	606/76	A61B 17/88
6083522	12/1999	Chu	N/A	N/A
6086593	12/1999	Bonutti	N/A	N/A
6086608	12/1999	Ek	N/A	N/A

6090072	12/1999	Kratoska	N/A	N/A
6099531	12/1999	Bonutti	N/A	N/A
6099537	12/1999	Sugai	N/A	N/A
6099550	12/1999	Yoon	N/A	N/A
6099552	12/1999	Adams	N/A	N/A
6102850	12/1999	Wang	N/A	N/A
6106545	12/1999	Egan	N/A	N/A
6117160	12/1999	Bonutti	N/A	N/A
6120536	12/1999	Ding	N/A	N/A
6123792	12/1999	Samida et al.	N/A	N/A
6125574	12/1999	Ganaja	N/A	N/A
6126677	12/1999	Ganaja	N/A	N/A
6139320	12/1999	Hahn	N/A	N/A
RE36974	12/1999	Bonutti	N/A	N/A
6149669	12/1999	Li	N/A	N/A
6152350	12/1999	Hayashi	N/A	N/A
6152949	12/1999	Bonutti	N/A	N/A
6155756	12/1999	Mericle	N/A	N/A
6159224	12/1999	Yoon	N/A	N/A
6159234	12/1999	Bonutti	N/A	N/A
6171307	12/2000	Orlich	N/A	N/A
6174324	12/2000	Egan	N/A	N/A
6179840	12/2000	Bowman	N/A	N/A
6179850	12/2000	Goradia	N/A	N/A
6187008	12/2000	Hamman	N/A	N/A
6190400	12/2000	Van de Moer	N/A	N/A
6190401	12/2000	Green	N/A	N/A
6200322	12/2000	Branch	N/A	N/A
6214049	12/2000	Gayer et al.	N/A	N/A
6217591	12/2000	Egan	N/A	N/A
6224593	12/2000	Ryan	N/A	N/A
6224630	12/2000	Bao	N/A	N/A
6228086	12/2000	Wahl	N/A	N/A
6231592	12/2000	Bonutti	N/A	N/A
6238395	12/2000	Bonutti	N/A	N/A
6238396	12/2000	Bonutti	N/A	N/A
6258091	12/2000	Sevrain	N/A	N/A
6264675	12/2000	Brotz	N/A	N/A
6267761	12/2000	Ryan	N/A	N/A
6273717	12/2000	Hahn	N/A	N/A
6280474	12/2000	Cassidy	N/A	N/A
6286746	12/2000	Egan	N/A	N/A
6287325	12/2000	Bonutti	N/A	N/A
6290644	12/2000	Green, II	600/209	A61B 17/0218
6293961	12/2000	Schwartz	N/A	N/A
6306159	12/2000	Schwartz	N/A	N/A
6309405	12/2000	Bonutti	N/A	N/A
6312448	12/2000	Bonutti	N/A	N/A
6324308	12/2000	Chen et al.	N/A	N/A

6328480	12/2000	Strike	N/A	N/A
6338730	12/2001	Bonutti	N/A	N/A
6340365	12/2001	Dittrich	N/A	N/A
6348056	12/2001	Bates	N/A	N/A
6358271	12/2001	Egan	N/A	N/A
6364897	12/2001	Bonutti	N/A	N/A
6364977	12/2001	Simon	N/A	N/A
6368325	12/2001	McKinley	N/A	N/A
6368343	12/2001	Bonutti	N/A	N/A
6371957	12/2001	Amrein	N/A	N/A
6409716	12/2001	Sahatjian et al.	N/A	N/A
6409742	12/2001	Fulton	N/A	N/A
6409743	12/2001	Fenton	N/A	N/A
6409768	12/2001	Tepic et al.	N/A	N/A
6419704	12/2001	Ferree	N/A	N/A
6423088	12/2001	Fenton	N/A	N/A
6425919	12/2001	Lambrecht	N/A	N/A
6428562	12/2001	Bonutti	N/A	N/A
6432115	12/2001	Mollenauer	N/A	N/A
6447516	12/2001	Bonutti	N/A	N/A
6450985	12/2001	Schoelling	N/A	N/A
6458228	12/2001	Yoshimoto	N/A	N/A
6461360	12/2001	Adam	N/A	N/A
6468293	12/2001	Bonutti	N/A	N/A
6475230	12/2001	Bonutti	N/A	N/A
6488196	12/2001	Fenton	N/A	N/A
6500195	12/2001	Bonutti	N/A	N/A
6503259	12/2002	Huxel	N/A	N/A
6527774	12/2002	Lieberman	N/A	N/A
6530933	12/2002	Yeung	N/A	N/A
6535764	12/2002	Imran	N/A	N/A
6544267	12/2002	Cole	N/A	N/A
6545390	12/2002	Hahn	N/A	N/A
6547792	12/2002	Tsuji	N/A	N/A
6551304	12/2002	Whalen	N/A	N/A
6554852	12/2002	Oberlander	N/A	N/A
6557426	12/2002	Reinemann	N/A	N/A
6558390	12/2002	Cragg	N/A	N/A
6568313	12/2002	Fukui	N/A	N/A
6569187	12/2002	Bonutti	N/A	N/A
6572635	12/2002	Bonutti	N/A	N/A
D477776	12/2002	Pontaoe	N/A	N/A
6585750	12/2002	Bonutti	N/A	N/A
6585764	12/2002	Wright	N/A	N/A
6592609	12/2002	Bonutti	N/A	N/A
6594517	12/2002	Nevo	N/A	N/A
6605090	12/2002	Trieu	N/A	N/A
6610080	12/2002	Morgan	N/A	N/A
6618910	12/2002	Pontaoe	N/A	N/A
6623486	12/2002	Weaver	N/A	N/A

6623487	12/2002	Goshert	606/76	A61B 17/68
6626944	12/2002	Taylor	N/A	N/A
6632245	12/2002	Kim	N/A	N/A
6635073	12/2002	Bonutti	N/A	N/A
6638279	12/2002	Bonutti	N/A	N/A
6641592	12/2002	Sauer	N/A	N/A
6645227	12/2002	Fallin	N/A	N/A
6652585	12/2002	Lang	N/A	N/A
6666877	12/2002	Morgan	N/A	N/A
6669705	12/2002	Westhaver	N/A	N/A
6679888	12/2003	Green	N/A	N/A
6685750	12/2003	Plos	N/A	N/A
6699240	12/2003	Fracischelli	N/A	N/A
6702821	12/2003	Bonutti	N/A	N/A
6705179	12/2003	Mohtasham	N/A	N/A
6709436	12/2003	Hover et al.	N/A	N/A
6709457	12/2003	Otte	N/A	N/A
6719765	12/2003	Bonutti	N/A	N/A
6719797	12/2003	Ferree	N/A	N/A
6722552	12/2003	Fenton	N/A	N/A
6733531	12/2003	Trieu	N/A	N/A
6764514	12/2003	Li	N/A	N/A
6770078	12/2003	Bonutti	N/A	N/A
6780198	12/2003	Gregoire	N/A	N/A
6786989	12/2003	Torriani	N/A	N/A
6796003	12/2003	Marvel	N/A	N/A
6818010	12/2003	Eichhorn	N/A	N/A
6823871	12/2003	Schmieding	N/A	N/A
6860885	12/2004	Bonutti	N/A	N/A
6878167	12/2004	Ferree	N/A	N/A
6890334	12/2004	Brace et al.	N/A	N/A
6893434	12/2004	Fenton et al.	N/A	N/A
6899722	12/2004	Bonutti	N/A	N/A
6913666	12/2004	Aeschlimann	411/908	B29C 66/1142
6916321	12/2004	TenHuisen	N/A	N/A
6921264	12/2004	Mayer	N/A	N/A
6923824	12/2004	Morgan	N/A	N/A
6932835	12/2004	Bonutti	N/A	N/A
6942684	12/2004	Bonutti	N/A	N/A
6944111	12/2004	Nakamura	N/A	N/A
6955540	12/2004	Mayer	N/A	N/A
6955683	12/2004	Bonutti	N/A	N/A
6958077	12/2004	Suddaby	N/A	N/A
6979376	12/2004	Nuss	N/A	N/A
6981983	12/2005	Rosenblatt	N/A	N/A
6997940	12/2005	Bonutti	N/A	N/A
7001411	12/2005	Dean	N/A	N/A
7004959	12/2005	Bonutti	N/A	N/A
7008226	12/2005	Mayer	N/A	N/A

7018380	12/2005	Cole	N/A	N/A
7033379	12/2005	Peterson	N/A	N/A
7048755	12/2005	Bonutti	N/A	N/A
7066960	12/2005	Dickman	N/A	N/A
7087073	12/2005	Bonutti	N/A	N/A
7090111	12/2005	Egan	N/A	N/A
7094251	12/2005	Bonutti	N/A	N/A
7104996	12/2005	Bonutti	N/A	N/A
7128761	12/2005	Kuras et al.	N/A	N/A
7128763	12/2005	Blatt	N/A	N/A
7147652	12/2005	Bonutti	N/A	N/A
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/581,719 filed Jan. 21, 2022, which is a continuation of U.S. patent application Ser. No. 15/224,823 filed Aug. 1, 2016, issued as U.S. Pat. No. 11,246,638 on Feb. 14, 2022, which is a continuation of U.S. patent application Ser. No. 12/711,540 filed Feb. 24, 2010, issued as U.S. Pat. No. 9,439,642 on Sep. 13, 2016, which is based upon and herein claims priority to U.S. Provisional No. 61/155,133 filed Feb. 24, 2009, and is a Continuation-in-Part of: U.S. patent application Ser. No. 12/202,210 filed Aug. 29, 2008, issued as U.S. Pat. No. 8,496,657 on Jul. 30, 2013; U.S. patent application Ser. No. 11/689,670 filed Mar. 22, 2007, issued as U.S. Pat. No. 9,610,073 on Apr. 4, 2017; U.S. patent application Ser. No. 11/671,556 filed Feb. 6, 2007, issued as U.S. Pat. No. 9,421,005 on Aug. 23, 2016; and U.S. patent application Ser. No. 11/416,618 filed May 3, 2006, issued as U.S. Pat. No. 7,967,820 on Jun. 28, 2011. The entire contents of each of the above are hereby expressly incorporated by reference into this disclosure as if set forth fully herein.

FIELD OF THE INVENTION

(1) The invention relates to the fastening and stabilizing of tissues, implants, and bondable materials, such as the fastening of a tissue and/or implant to a bondable material, the fastening of an implant to tissue, and/or the fastening of an implant to another implant. This may involve using an energy source to bond and/or mechanically stabilize a tissue, an implant, a bondable material, and/or other biocompatible material. The present invention also relates to the use of an energy source to remove and/or install an implant and/or bondable material or to facilitate solidification and/or polymerization of bondable material.

BACKGROUND OF THE INVENTION

(2) Body tissue often requires repair and stabilization to address weak or fractured bone, torn ligament or tendon, ripped muscle, or separation of soft tissue from bone. There are numerous methods to facilitate this repair and stabilization. For example, weak or fractured bones can be reinforced with bondable material, i.e. bone cement. Over time, these bondable materials may loosen due to tissue deterioration, improper installation of bondable materials, or deterioration of the bondable materials over potentially damaging to the tissue. After the bondable material is sufficiently removed using conventional methods, bondable material is reapplied to the tissue. Therefore, previous stabilization methods provided for the reapplication of bondable materials and did not utilize existing bondable materials. There is a need for an improved method to utilize existing bondable materials to stabilize tissue and implants.

- (3) In another example, bondable materials are used for the installation of implants, i.e. example bone cement. However, some implants loose stability over time. Previous stabilization methods require removal of the implant and the remaining bondable material left on the bone. After the bondable materials are removed, new bondable material is applied to the implant and/or bone. Again, this is a time consuming process, potentially damaging the surrounding tissue during the removal of the implant and remaining bondable material.
- (4) Bone plates may be positioned internal to the skin, i.e. positioned against the fractured bone, or may be positioned external to the skin with rods connecting the bone and plate. Conventional bone plates are particularly well-suited to promote healing of the fracture by compressing the fracture ends together and drawing the bone into close apposition with other fragments and the bone plate. However, one drawback with plates and screws is that with the dynamic loading placed on the plate, loosening of the screws, and loss of stored compression can result. There is a need for additional fixation devices and methods related to bone plates and other implants providing support to bone.
- (5) In addition to internal or external bone plates, surgeons sometimes use intramedullary rods to repair long bone fractures, such as fractures of the femur, radius, ulna, humerus, fibula, and tibia. The rod or nail is inserted into the medullary canal of the bone and affixed therein by screws or bolts. After complete healing of the bone at the fracture site, the rod may be removed through a hole drilled in the end of the bone. One problem associated with the use of today's intramedullary rods is that it is often difficult to treat fractures at the end of the long bone. Fastener members, such as bolts, are positioned through the cortical bone and into threaded openings in the rod. However, the number and positioning of the bolt/screw openings are limited at the tip of the rod because of the decreased surface area of the rod and the reduced strength at the tip of the rod. Fractured bone sections at the distal end of a femur, for example, may not be properly fastened to using conventional intramedullary rod stabilization techniques. Therefore, additional fixation devices and methods are required for use with intramedullary rods.
- (6) Other common methods to address weak or fractured bones use a combination of bone screws, bone plates, and intramedullary rods. Conventional methods of using bone screws required a sufficient depth within the bone to stabilize a bone plate. However, weak or fracture bones have limited purchase, as portions of the bone may be unfit for the use of bone screws. Furthermore, if an intramedullary rod has been used to stabilize the bone, the fixation area is further limited as surgeons generally avoid tapping into areas of bone with an underlying intramedullary rod. An improved method of stabilizing existing bone plates and intramedullary rods is needed.
- (7) Existing systems and techniques for repairing tissue, like the ones previously described, can be complex, time consuming, lack the characteristic of being employed with precision, be damaging to tissue, and/or fail to provide a robust fastening of tissue. Therefore, there is a need for an apparatus and method for the fastening of tissue that involves a reduction in completion time, greater strength and precision, utilization of previously implanted materials, and preservation of living tissue. There is a need for a system that utilizes of previously installed fixation devices and techniques.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:
- (2) FIG. 1 is a perspective view of an exemplary vibratory energy device;
- (3) FIG. 2 illustrates an embedding implant and end effector of the invention;
- (4) FIG. 3 illustrates an alternative view of FIG. 2;

(5) FIG. 4 illustrates an embedding implant connected to an end effector and embedded in a bondable material;

(6) FIG. 5 illustrates an alternative welding horn or embedding implant of the invention;

(7) FIG. 6 illustrates an alternative configuration of the end effector of FIG. 3;

(8) FIG. 7 illustrates a cross section through the center of a long axis of the end effector of FIG. 6;

(9) FIG. 8 illustrates a cross section through the center of a long axis of a fastening implant;

(10) FIG. 9 illustrates an alternative view of FIG. 8;

(11) FIG. 10 illustrates the embedding implant embedded in bondable material and engaged with a fastening implant;

(12) FIG. 11 illustrates an alternative configuration of the fastening implant of FIG. 9;

(13) FIG. 12 illustrates a cross section through the center of a long axis of the fastening implant of FIG. 11;

(14) FIG. 13 illustrates a washer for use with implants of the invention;

(15) FIG. 14 illustrates a cross section through the center of a long axis of FIG. 13;

(16) FIG. 15 illustrates embedding implant embedded in bondable material;

(17) FIG. 16 illustrates the fastening implant of FIG. 11 disposed in the washer of FIG. 13 and engaged with the embedding implant of FIG. 15;

(18) FIG. 17 illustrates the fastener of FIG. 11 bonded and/or staked to the washer of FIG. 13;

(19) FIG. 18 illustrates alternative configurations of implant of the invention;

(20) FIG. 19 illustrates alternative configurations of FIG. 18;

(21) FIG. 20 illustrates an alternative configuration of an embedding implant;

(22) FIG. 21 illustrates a cross section through the center of a long axis of FIG. 20;

(23) FIG. 22 illustrates an alternative configuration of the embedding implant of FIG. 2;

(24) FIG. 23 illustrates a cross section through the center of a long axis of FIG. 22;

(25) FIG. 24 illustrates an alternative configuration of the embedding implant of FIG. 2;

(26) FIG. 25 illustrates a cross section through the center of a long axis of FIG. 24;

(27) FIG. 26 illustrates the use of a fastening implant and an embedding implant to secure a supporting implant;

(28) FIG. 27 illustrates the assembled configuration of FIG. 26;

(29) FIG. 28 illustrates devices and methods related to utilization of implants and bondable materials;

(30) FIG. 29 illustrates alternatives for the devices and methods of FIG. 28;

(31) FIG. 30 illustrates alternatives for the devices and methods of FIG. 29 and illustrates a method of connecting an instrument to an implant;

(32) FIG. 31 illustrates an alternative method of connecting an instrument to an implant;

(33) FIG. 32 illustrates the affixing an attachment to an implant;

(34) FIG. 33 illustrates the use of a reservoir.

(35) FIG. 34 illustrates an alternative configuration of FIG. 33.

(36) FIG. 35 illustrates alternative types of reservoirs in an alternative type of implant.

(37) FIG. 36 illustrates an additional alternative type of implant.

(38) FIG. 37 illustrates fixation to previously implanted bondable material.

(39) FIG. 38 illustrates an alternative view of FIG. 37.

(40) FIG. 39 illustrates an alternative configuration of the instrument of FIG. 1.

(41) FIG. 40 illustrates an exploded view of the instrument of FIG. 39.

(42) FIG. 41 illustrates a method of utilization for the instrument of FIG. 39.

(43) FIG. 42 illustrates an alternative configuration of a fastening implant.

(44) FIG. 43 illustrates an alternative view of FIG. 42.

(45) FIG. 44 illustrates an alternative configuration of an embedding implant.

(46) FIG. 45 illustrates an alternative view of FIG. 44.

(47) FIG. 46 illustrates an alternative configuration of a fastening implant.

- (48) FIG. 47 illustrates an alternative view of FIG. 46.
- (49) FIG. 48 illustrates an energy signal through an end effector and implant.
- (50) FIG. 49 illustrates an alternative configuration of an energy signal through an end effector and implant.
- (51) FIG. 50 illustrates an exploded view of the fastening implant of FIG. 41.
- (52) FIG. 51 illustrates the fastening implant of FIG. 50 after staking.
- (53) FIG. 52 illustrates an exploded view of an alternative type of fastening implant.
- (54) FIG. 53 illustrates an alternative view of FIG. 52.
- (55) FIG. 54 illustrates an alternative configuration of a fastening implant.
- (56) FIG. 55 illustrates the fastening implant of FIG. 54 bonded to another implant.
- (57) FIG. 56 illustrates an alternative configuration of a fastening implant or embedding implant.
- (58) FIG. 57 illustrates an expandable configuration of a fastening implant and expanding implant.
- (59) FIG. 58 illustrates an alternative view of FIG. 57.
- (60) FIG. 59 illustrates an alternative configuration of an implant, interference implant, and fastening implant.

SUMMARY

- (61) As is described in further detail below, fasteners may be embedded within solidified bondable material, for example a grouting agent such as bone cement (PMMA) or other acrylic based material. In an embodiment in accordance with the invention, an embedding fastener may be connected to an end effector of a vibratory energy generator. The embedding fastener may be adapted to enter and engage the bondable material that has been locally melted by vibratory energy, and to be securely retained therein once the bondable material has cooled and hardened.
- (62) The end effector may be provided in any of a variety of shapes, one example being an elongated rod or shaft, connectable to a hand piece at a proximal end, and operative to transmit vibratory energy at a distal end. The fastener may be adapted to connect to the distal end of the end effector, for example by mechanical interlocking, threading, twist lock configurations, friction fitting, or adhesive attachment. The mechanical connection must be operative, however, to communicate the vibratory energy from the end effector to the fastener.
- (63) The fastener is adapted to be securely retained within the bondable material or adhesive, in one embodiment, by being provided with a shaped or contoured surface upon which the adhesive may grip once hardened. A roughened or porous surface may be provided alone or in combination with a shaped surface to increase purchase in bondable material and/or facilitate an interference fit.
- (64) The fastener may further be provided with a taper at a leading end which first enters the adhesive. The taper improves performance, at least, by promoting accurate tracking and movement of the fastener into the adhesive, piercing tissue, and facilitating initial melting by concentrating vibratory energy over a smaller surface region.
- (65) In a further embodiment of the invention, the embedded bone cement fastener (also referred to as an embedding fastener or embedding implant) described above is provided with one or more radial gaps, chambers, or ports, extending from a central bore. A polymeric fastener is inserted within the central bore, and vibratory energy is applied to the polymeric fastener, whereby polymer at the interface between the embedded fastener and the polymeric fastener melts. When the polymer melts, and particularly as pressure is applied to the polymeric fastener in the direction of insertion, polymer enters the ports, flowing in a direction away from the central bore. When vibratory energy is discontinued, the polymer solidifies, and the polymer fastener is thereafter secured within the embedding fastener.
- (66) The embodiments of the present invention may be utilized with limitless fixation techniques and in conjunction with other fasteners and implants. Furthermore, the embodiments herein may assist in the installation and removal of tissue and implants. Moreover, the embodiments of the present invention may assist in the delivery of therapeutic agents and employ methods that facilitate tissue growth and repair. In addition, the embodiments herein may be used to apply

vibratory energy to remove and/or install an implant in bondable material or to facilitate solidification and/or polymerization of the bondable material.

DETAILED DESCRIPTION

(67) The invention relates to the devices and methods for the utilization of bondable materials and bondable materials, fixation and fastening of tissue to tissue, an implant to tissue, and an implant to an implant both inside and outside the body. The invention additionally relates to removing and anchoring implants to bondable materials and/or other biocompatible materials, anchoring implants using previously implanted and hardened bondable materials, and fixation using vibratory energy, mixing, solidifying, bonding, and/or mechanical interlocking of materials. The present invention also relates to the use of an energy source to install and/or remove an implant or bondable material or to facilitate the solidification and/or polymerization of a bondable material

(68) The methods and devices disclosed herein may be used in conjunction with any medical procedure on the body. The stabilization, fastening, and/or repair of tissue or an implant may be performed in connection with any medical procedure related to a joint, bone, muscle, ligament, tendon, cartilage, capsule, organ, skin, nerve, vessel, or other body parts. For example, tissue may be stabilized during intervertebral disc surgery, kyphoplasty, knee surgery, hip surgery, organ transplant surgery, bariatric surgery, spinal surgery, anterior cruciate ligament (ACL) surgery, tendon-ligament surgery, rotator cuff surgery, capsule repair surgery, fractured bone surgery, pelvic fracture surgery, avulsion fragment surgery, shoulder surgery, hernia repair surgery, and surgery of an intrasubstance ligament tear, annulus fibrosis, fascia lata, flexor tendons, etc.

(69) Also, an implant may be inserted within the body and fastened to tissue with the present invention. Such implant insertion procedures include, but are not limited to, partial or total knee replacement surgery, hip replacement surgery, shoulder replacement surgery, bone fastening surgery, etc. The implant may be an organ, partial organ grafts, tissue graft material (autogenic, allogenic, xenogenic, or synthetic), collagen, a malleable implant like a sponge, mesh, bag/sac/pouch, collagen, or gelatin, or a rigid implant made of metal (porous or nonporous), polymer, composite, or ceramic. Collagen may provide the benefit of bolstering tissue growth. Additionally, a desiccated collagen may be used to absorb surrounding fluid, which may provide the additional benefit of applying pressure on the tissue being repaired. Other implants include breast implants, biodegradable plates, porcine or bovine patches, metallic fasteners, compliant bearing for medial compartment of the knee, nucleus pulposus prosthetic, stent, suture, suture anchor, tissue graft, tissue scaffold, biodegradable collagen scaffold, and polymeric or other biocompatible scaffold. The scaffold may include fetal cells, stem cells, embryonic cells, enzymes, and proteins.

(70) In this application, the term “bondable” or “bondable material” is used to refer to the materials discussed herein, as well as any material, suitable for in vivo applications, which can be softened and made flowable by the application of heat (such as heat produced with vibratory energy such as ultrasonic energy), and which, when softened, may become tacky and will bond to other materials and will flow to fill available space. Thus, the material may be thermoplastic, but it may also exhibit tackiness or bonding ability when in its plastic form. Many materials suitable for in vivo applications are made of or incorporate such bondable materials. Generally speaking, the amount of heat needed to soften and make flowable should be within a temperature range that does not produce substantial thermal tissue necrosis. Alternatively stated, the amount of heat required to soften the bondable material during vibratory bonding is substantially confinable, due to the thermal properties of the bondable material, to an area of contact between the objects which are being bonded, thereby protecting living body tissue near the contact between the two objects from substantial thermal tissue necrosis. Any embodiment herein may be used with any of the materials and/or applications disclosed herein or known in the art.

(71) The fixation and fastening system and other embodiments of the present invention contemplates the use any materials that may include or be used in conjunction with bondable

materials for bonding and/or staking within the human body. Implants that may be used as fasteners may also be referred to as fastening implants. Bondable material used may include, but are not limited to, biocompatible, degradable, biodegradable, bioerodible, bioabsorbable, mechanically expandable, hydrophilic, bendable, deformable, malleable, riveting, threaded, toggling, barded, bubbled, laminated, coated, blocking, pneumatic, one-piece, multi-component, solid, hollow, polygon-shaped, pointed, self-introducing, and combinations thereof. Also, the devices may include, but are not limited to, metallic material, polymeric material, ceramic material, composite material, body tissue, synthetic tissue, hydrophilic material, expandable material, compressible material, bondable material, and combinations thereof. Bondable material may also include polymethyl methacrylate (known as “PMMA” or “bone cement”), glue, adhesive, and/or other grouting agents or acrylic materials used for fixation.

(72) In this application, “bond”, “bonded”, and “bonding” includes, but is not limited to, attaching, engaging, connecting, binding, adhering, and/or fastening one or more materials through resistive heating, mechanical interlocking, application of force, application of grouting agents (i.e. bone cement), adhesives and/or solvents, spraying, radiofrequency, vibratory energy (i.e. ultrasound), microwave, laser, electromagnetic, electro shockwave therapy, plasma energy (hot or cold), and other suitable method described herein or known in the art.

(73) Preferably, materials of the present invention can melt with the application of energy, becoming gel-like, tacky, and/or soft. The energy source and the technique used to bond and/or stake the material within the body can be selected to minimize or avoid damage to surrounding body tissue. Exemplary materials that may be used may include polymers, ceramics, composites, and metals, although other materials may also be suitable for use with the invention. While the present invention contemplates the use of any of these materials in any of the following embodiments, polymeric material is used in the following examples and description simply to illustrate how the invention may be used.

(74) There are a limitless number of materials may be used for the present invention. Examples of amorphous polymers are polycarbonate (LEXAN), polystyrene, polysulfone (ULDALL), and acrylics polycarbonate (ABS and styrenes). Examples of semi-crystalline polymers include acetyl (DELFIN), nylon, polyester, polyethylene, polyether ether ketone, polypropylene, polyvinylchloride (PVC), and Caprolactam. Biodegradable semi-crystalline polymers may include polylactic acid and polyglycolic acid. Copolymers of PGA and PLA may also be used. Poly-L-lactide (PLLA) or other forms of PLA may also be used. Other polymers which may be used with the present invention, either as a thermoplastic or non-thermoplastic, are polyethylene glycol (PEG)-copolymers and D,L-lactide-co-glycolide polyesters. Some semi-crystalline materials are particularly suitable for surgical bonding and/or staking, especially vibratory bonding and staking. Examples of such materials include PAEK (polyaryletherketone), including PEEK (polyetheretherketone) and PEKK (polyetherketoneketone).

(75) In addition to PEEK and the other polymers described herein, the implants, devices, and methods of the present invention may use keratin, a naturally occurring polymer. Keratin may be vibratory bonded or staked to itself, to other implants, or within tissue. This may be performed in the operating room or intracorporeally. Keratin may be bonded to collagen or to other known polymers. In an exemplary application, keratin may be used to fasten tissue to bone since keratin has BMP and tissue scaffold properties. It is contemplated that any of devices and methods disclosed herein may utilize keratin alone or in combination with PEEK, polylactic acid, or other polymer. Keratin may be used to make fasteners, disc replacements, joint replacement components, stents, cell scaffolds, drug reservoirs, etc. Also, joint bearing surfaces may include keratin with or without collagen or chondrocytes. The bearing surfaces may be fastened to a joint component using PEEK or PLA fasteners.

(76) Another polymer that can be used with the present invention is a class of natural materials, called polyhydroxyalkanoates, or PHA polymers. These polymers are synthesized in nature by

numerous microorganisms, and they have been recently recognized as the fifth class of naturally occurring biopolymers (along with the polyamino acids, polynucleic acids, polysaccharides, and polyisoprenoids). Unlike the other naturally occurring biological polymers, however, the PHA polymers are thermoplastic, i.e. they may be repeatedly softened with heat and hardened with cooling. As such, these polymers can be processed much like other plastics. A specific example of a PHA polymer that could be used is poly-4-hydroxybutyrate material. Such PHA polymers are available from Tephra Inc of Lexington, MA.

(77) Fasteners of the invention may utilize or be coated with polymethylmethacrylate (PMMA), in order to promote bonding with PMMA used in the body, or PMMA could be incorporated into polymer of the fastener, or deposited within cavities or shapes formed in the fastener surface, including threaded, roughened, porous, or nano textures. A fastener may be thus coated with PMMA, or formed entirely of PMMA, and may be heat bonded, advantageously using ultrasound, to another PMMA surface or other adhesive surface, otherwise as described herein with respect to bone cement. Although PMMA, known generally as bone cement, and other polymers may function more as a grouting agent than a cement or adhesive. The term “bondable material” is used throughout the specification for simplicity.

(78) In accordance with the invention, metals are advantageously connected with fasteners incorporating polymeric materials. Any of a variety of metals may be used, either smooth or formed with at least portions formed of metal, or a roughened or porous surface, or formed with cavities or other shapes upon which polymeric material may mold, enter, adhere, or otherwise affix. The polymer is softened in accordance with the invention through the application of heat, including heat created using vibratory energy, to become tacky, or sufficiently softened in order to bond on a microscopic level, or a macroscopic level through adaptation to the surface structure of the metal. For use in vivo, biocompatible metals are used, including stainless steel, nitinol or other SMA (shape metal alloy), tantalum, porous tantalum, titanium, cobalt-chrome alloys, and other metals such as are known to those skilled in the art. Additional related information, including bonding polymers and metals, and polymer to polymer bonding of implant materials, may be found in U.S. Pat. No. 5,163,960 entitled “Surgical devices assembled using bondable materials”, and U.S. Pat. No. 7,104,996 entitled “Method of performing surgery”, the contents of each of which being incorporated herein by reference.

(79) The fastening device of the present invention may include therapeutic substances to promote healing. These substances could include antibiotics, hydroxyapatite, anti-inflammatory agents, steroids, antibiotics, analgesic agents, chemotherapeutic agents, bone morphogenetic protein (BMP), demineralized bone matrix, collagen, growth factors, autogenous bone marrow, progenitor cells, calcium sulfate, immunosuppressants, fibrin, osteoinductive materials, apatite compositions, germicides, fetal cells, stem cells, enzymes, proteins, hormones, cell therapy substances, gene therapy substances, and combinations thereof. These therapeutic substances may be combined with the materials used to make the device. Alternatively, the therapeutic substances may be impregnated or coated on the device. Time-released therapeutic substances and drugs may also be incorporated into or coated on the surface of the device. The therapeutic substances may also be placed in a bioabsorbable, degradable, or biodegradable polymer layer or layers.

(80) The therapeutic agents may also be placed within one or more cavities disposed in a fastening device of the present invention. Different agents may be disposed in different cavities of the device to specifically tailor the implant for a particular patient. Dosages of the therapeutic agent may be the same or different within each of cavities as well. The cavities may include a cover which may release the agent in a controlled or timed manner. The cover may be biodegradable or bioerodible to allow the agent to release to surrounding tissue. Examples of suitable therapeutic agents include bone growth inducing material, bone morphogenic proteins, osteoinductive materials, apatite compositions with collagen, demineralized bone powder, or any agent previously listed. U.S. patent application Ser. No. 11/549,994 entitled “Drug Eluting Implant” discloses means for delivering

therapeutic agents. The above-mentioned patent application is incorporated by reference herein in its entirety.

(81) The fastening devices of this and other embodiments of the invention may be used in combination with fasteners in the prior art. Examples of fasteners, implants, and their methods of employment may be found in U.S. Pat. Nos. 5,163,960; 5,403,348; 5,441,538; 5,464,426; 5,549,630; 5,593,425; 5,713,921; 5,718,717; 5,782,862; 5,814,072; 5,814,073; 5,845,645; 5,921,986; 5,948,002; 6,010,525; 6,045,551; 6,086,593; 6,099,531; 6,159,234; 6,368,343; 6,447,516; 6,475,230; 6,592,609; 6,635,073; and 6,719,765. Other fastener types are disclosed in U.S. patent application Ser. Nos. 12/202,210; 10/102,413; 10/228,855; 10/779,978; 10/780,444; and 10/797,685. The above cited patents and patent applications are hereby incorporated by reference in their entirety.

(82) With reference to FIG. 1, any known energy emitting instrument may be used with the surgical system of the present invention. Instrument **100** may produce energy such as resistive heating, radiofrequency, ultrasound (vibratory), microwave, laser, electromagnetic, electro shockwave therapy, plasma energy (hot or cold), and other suitable energy disclosed herein or known in the art. FIG. 1 illustrates an exemplary handpiece or instrument **100** that may be used with the present invention. The instrument **100** may be a vibratory energy generator with a sheath **102** to cover and protect the end effector **104** and engage a fastener/implant near engagement feature **106**. As will be discussed below, the instrument may be used to bond and/or mechanically interlock fasteners and other embodiments of the present invention. Additional embodiments of instrument **100** are disclosed in U.S. patent application Ser. No. 12/202,210 entitled “Methods and Devices for Utilizing Thermal Energy to Bond, Stake and/or Remove Implants”, which is incorporated by reference herein.

(83) With reference to FIGS. 2-7, end effector **104** may be utilized with anchor or embedding fastener **110**. Embedding fastener **110** may also be referred to as an embedding implant. FIG. 2 illustrates end effector **104** that connects to embedding fastener **110** with distal end **108**. The connection between distal end **108** and embedding fastener **110** may utilize threads, magnetism, friction, taper, ball and socket, linkage, adhesive, interlocking shapes, and other connections known in the art. Additionally, distal end **108** and embedding fastener **110** may be permanently or detachably connected.

(84) As shown in FIG. 2, embedding fastener **110** may further be provided with a taper **114**, which first enters bondable material **120** in FIG. 4. Taper **114** may improve performance, for example, by promoting accurate tracking and movement of embedding fastener **110** into bondable material **120**, piercing body tissue, and facilitating initial melting by concentrating vibratory energy over a smaller surface region. Although embedding fastener **110** may be made of any material described herein or known in the art, it may be preferable to use titanium.

(85) As discussed in further detail below, embedding fastener **110** may also have feature **112** and/or feature **116**, either or both may be a surface feature, recess, or pass through a portion or the entirety of embedding fastener **110**.

(86) Referring to FIG. 3, an embodiment of end effector **104** may have proximal end **118** to the other components of instrument **100**. The connection between instrument **100** and proximal end **118** may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other methods disclosed herein or known in the art.

(87) As shown in FIGS. 3-7, end effector **104** may be provided in any of a variety of shapes, one example being an elongated rod or shaft, connectable to a hand piece at a proximal end **118**, and operative to transmit vibratory energy at a distal end **108**. While a rod shape is shown and selected for reduced manufacturing cost, end effector **104** may have the form of box or hex channel, oval or other shape, provided it communicates vibratory energy to a distal end **108**, an attached fastener, or embedding fastener **110**. Additionally, feature **111** of FIG. 5 may be used on embedding fastener **110** or integrated into end effector **104** (not shown).

(88) In an additional embodiment, embedding fastener **110** is adapted to connect to distal end **108**

of end effector **104** by mechanical interlocking, as by a bore in embedding fastener **110**, sized to receive distal end **108** of end effector **104**, optionally provided with internal or external threading (not shown), wherein post **108** has mating threads. Additionally, the connection may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other methods disclosed herein or known in the art. Similarly, a bore or aperture may be provided in end effector **104**, mateable with a post or projection on embedding fastener **110**. Other mechanical connections are contemplated, including twist lock configurations, friction fitting, or adhesive attachment. The mechanical connection should preferably be operative to communicate vibratory energy from end effector **104** to embedding fastener **110**, as by a firm mechanical connection.

(89) As shown in an embodiment of FIG. 4, embedding fastener **110** may be adapted to be securely retained within bondable material **120** by being provided with a shaped or contoured surface upon which the softened bondable may adhere. A roughened or porous surface may be provided alone or in combination with shaped surface thereby providing for increased purchase in bondable material **120**.

(90) With reference to FIG. 4, embedding fastener **110** may be embedded within solidified bone bondable material **120**, for example PMMA, acrylic based adhesive, or other bondable materials. In the present invention, embedding fastener **110** is connected to end effector **104** of an embodiment of instrument **100**, such as a vibratory energy generator as shown in FIG. 1. Embedding fastener **110** is adapted to enter and engage bondable material **120** or bondable material **120** that has been locally melted by vibratory energy (as shown in FIG. 4), through contact between embedding fastener **110** and bondable material **120** during operation of instrument **100**. Embedding fastener **110** is securely retained by bondable material **120** once the latter has hardened. Although the embodiment in FIG. 4 may be used under a limitless number of configurations and settings, Table 1 is being set forth with operative examples:

(91) TABLE-US-00001 TABLE 1 Titanium Embedding Fastener Bonded into PMMA Embedding Fastener Type: Shown in FIG. 20 Instrument: Handpiece SN0105 with tuning of 39,000-45,000 Hz System Settings: Sample 1: 40,850 Hz, 100W, 2.0 sec weld time Sample 2: 40,750 Hz, 75 W, 1.5 sec weld time Sample 3: 40,800 Hz, 75 W, 1.0 sec weld time Sample 4: 40,750 Hz, 75 W, 1.0 sec weld time Test Force Sample Power Time Applied to Deformation Number (watts) (sec) Break (lbs.) Depth (inches) 1 63 2.56 46.7 0.116 2 48 2.01 61.9 0.119 3 48 1.58 32.5 0.109 4 48 1.47 31.5 0.098

(92) Once anchored, end effector **104** and embedding fastener **110**, embedded in bondable material **120**, may remain connected. Alternatively, end effector **104** may be removed and another fastener of a similar or different design may be connected to an implanted embedding fastener **110** as shown in FIGS. 8-10. In a further embodiment, fastener **124** such as described in the incorporated patents and applications may be fastened to an implanted or installed embedding fastener **110**. Fastener **124** may have fastener bore **128** as shown in FIGS. 8-9. Referring to FIG. 10, a bondable insert **134** may be secured into fastener bore **128**. Bondable insert **134** may be secured by press fitting, threading, or bonding to fastener bore **128** and/or embedding fastener **110**. The fastener **124** may be utilized as detailed in U.S. patent application Ser. No. 12/202,210, which has been incorporated by reference herein. In further embodiments, any fastener described in the related references cited in paragraph [0001] or discussed herein may be fastened to the embedding fastener **110**, then secured in its respective manner.

(93) In an additional embodiment, embedding fastener **110** may be used to remove an implant and/or bondable material **120**. For example, the ability of conventional medical tools to remove a previously installed implant or bone cement may be limited. Embedding fastener **110** may be used to obtain additional fixation. Once embedding fastener **100** is secured to the implant and/or bondable material **120**, force and/or vibratory energy may be used to remove the implant and/or bondable material **120**.

(94) FIGS. 11-17 show an additional embodiment for use with an implanted embedding fastener

110. Fastener **136** is shown in FIGS. **11-12** and washer **146** is shown in FIGS. **13-14**. Although fastener **136** and washer **146** may be made of any material disclosed herein or known in the art, it may be preferable to use PEEK. After embedding fastener **110** has been secured with respect to bondable material **120**, fastener **136** may be engaged into embedding fastener **110**. In another embodiment, washer **146** may be used in conjunction with fastener **136** as shown in FIGS. **15-17**. Additionally, fastener **136** may be bonded to embedding fastener **110** and/or washer **146**.

(95) Referring to FIGS. **18-19**, fastener **136**, washer **146**, and/or tissue implant **154** may be used to secure soft tissue **152** to hard tissue **156**, for example to secure the rotator cuff tissue to the proximal humerus or for any other procedure disclosed herein. Additionally, washer **146** and/or tissue implant **154** could be made of collagen or other materials that promote tissue growth.

(96) With reference to FIGS. **20-25**, embedding fastener **110** may be provided with channel **112**. For example, channel **112A**, **112B**, **112C**, and/or **116** may be used. Channel **112** may extend through the surface of embedding fastener **110** to facilitate the bonding of embedding fastener **110** to bondable material **120**, fastener **136**, and/or any fastener disclosed herein. Channel **112** may provide a path for softened and/or molten bondable material to be displaced, providing room for entry of embedding fastener **110**. Channel **116** may also include radial gaps, chambers, or ports. To accommodate for embedding fastener **110** displacing a substantial amount of material, channels may be extended along the entire length of embedding fastener **110**, and may further extend along end effector **104**. Channel **116** may be further operative to reduce the possibility of rotation of fastener **110** within bondable material **120**. Channel **116** is thus disposed to extend into bondable material **120** after insertion, and may extend to the face of embedding fastener **110**. Additionally, embedding fastener **110** may have feature **158** to help attach and remove it from end effector **104**.

(97) In an embodiment shown in FIGS. **26-27**, implant **162** may be coated in bondable material **164** and implanted in body tissue **160**. For example, a metal rod coated with bone cement may be placed in the intramedullary canal of a bone. In an embodiment, support **168** may be placed in a location to facilitate stabilization. Support **168** and/or washer **146** may be referred to as a supporting implant. One or more holes may be formed in body tissue **160** and up to or into bondable material **164** to coincide with the holes in support **168**. Embedding fastener **110** may be placed through the holes in body tissue **160** and secured to and/or bonded to bondable material **164** as discussed herein. Then, one or more of fastener **136** are secured to and/or bonded to the one or more embedding fastener **110**, thereby securing support **168** relative to body tissue **160**.

(98) Referring to FIG. **28**, implant **162** may be installed in a body tissue **160** with bondable material **164**, for example bone cement. Bondable material **164** may be any material described herein or known in the art. Implant **162** may require stabilization because implant **162** has become loose and/or requires stabilization due to tissue defect **166**, for example a periprosthetic fracture. Tissue defect **166** may include, but is not limited to, damaged, deformed, and/or diseased bone, muscle, ligament, tendon, cartilage, capsule, organ, skin, nerve, vessel, or other body part. For example, a femur may be fractured or contain osteoporosis. Support **168** is fixed to body tissue **160** with fastener **170** to provide stabilization. Support **168** may be an internal bone plate, an external bone plate, a spinal plate, a wedge, a cushion, a pad, or other biocompatible support used for stabilization of tissue and/or implants. Fastener **170** may be any fastener described herein or any other biocompatible fastener known in the art.

(99) In an embodiment, implant **162** has been previously installed and requires stabilization. One or more holes are formed through body tissue **160** and up to or into bondable material **164**.

Embedding fastener **110** is inserted through a hole and bonded to bondable material **164** by utilizing instrument **100** described above. Fastener **170** engages embedding fastener **110** to secure support **168** to body tissue **160**. The head of fastener **170** may be deformed and/or bonded to support **168** to reduce loosening of fastener **170**.

(100) Embedding fastener **110** can also be bonded to bondable material **164C/D** that is within or on the surface of body tissue **160**. For example, bondable material may have been used to repair tissue

defect **166**. Bondable material **164** may be within or on the surface of body tissue **160**. A hole is formed up to or into the bondable material **164**. Embedding fastener **110** is bonded into bondable material **164**. Fastener **170** passes through support **168** and into engagement with embedding fastener **110** to secure support **168** relative to body tissue **160**.

(101) In another embodiment, embedding fastener **110A/B** can be bonded to and/or into implant **162**. The procedure is performed as described above, except the embedding fastener **110** may be bonded directly to implant **162**.

(102) In another embodiment, bondable material **164** may asymmetrically cover all or a portion of implant **162**. The thickness of bondable material **164** could vary in the radial direction or along the length of implant **162**. An asymmetrically coated implant **162** may provide additional purchase for fastener **136** or indication of orientation or position of implant **162**.

(103) In an additional embodiment, indirect visualization may be used to identify and/or change the orientation or position of implant **162** or fastener **136**. Examples of indirect visualization may include endoscopic guidance, computer assisted navigation, magnetic resonance imaging (MM), CT scan, ultrasound, fluoroscopy, X-ray, or other visualization technique disclosed in any of the references incorporated herein. Asymmetric coating, radiopaque markers, or other features identifiable with indirect visualization may be used to identify and/or adjust orientation or position. Indirect visualization may also be used to align fastener **136** with holes in implant **162** or bondable material **164**. The holes may be predrilled in implant **162** or bondable material **164** or may be drilled after installation of **162**. Indirect visualization may be used to create a hole or holes in tissue to align with holes in implant **162** or bondable material **164**.

(104) For example, an intramedullary rod could be asymmetrically coated with PEEK. The intramedullary rod could have predrilled holes in the PEEK coating. After the rod is installed in the intramedullary canal of the tibia, the orientation of the rod may be determined using indirect visualization to locate the area with a thicker coating. The orientation of the rod may be adjusted to the appropriate location for holes to be made through the tissue.

(105) With further reference to FIG. **28-29**, various types of fastening devices are used to position support **168** along body tissue **160**. Alternatively, support **168** may be positioned upon the surface of the skin, or at any point between the tissue surface and the skin, according to the requirements of the surgical procedure. Further, support **168** may be placed within the bone, for example in an intramedullary canal.

(106) Referring to FIG. **29**, fastener **170** may be used in intramedullary, percutaneous, and/or retrograde approaches. Fastener **170** may be bonded to bondable material **164**, or a surface of implant **162**. The head of fastener **170** may be provided, or may be formed using vibratory energy. A head may also be formed on the distal end of fastener **170**. Fasteners **170E** are shown to be bonded into the bondable material **164**. Fastener **170F** is shown to be bonded at the distal end and/or to bondable material **164** within the body tissue and is placed through tissue defect **166**. Fasteners **170G** are shown passing directly through body tissue **160**, which may be fastener **170T** and sleeve **171T** in FIGS. **52-53** and as described below. Additional embodiments of fastener **170** are disclosed in U.S. patent application Ser. No. 12/202,210 entitled "Methods and Devices for Utilizing Thermal Energy to Bond, Stake and/or Remove Implants", which is incorporated by reference herein.

(107) Additionally, cerclage wire **172** may be employed as known in the art, to provide further stabilization, in combination with fastener **170**. For example, cerclage wire **172A** may be bonded to support **168**. Bondable material **164E** could be used to affix cerclage wire **172A** to support **168**. In another example, cerclage wire **172B** may be tied around support **168**. Also, cerclage wire **172C** may be fastened using a mechanical or bonded crimp **174**. In additional example, cerclage wire **172D** may be fastened to the side of support **168** or between support **168** and body tissue **160**.

(108) Referring to FIG. **30**, end effector **104** of instrument **100** may be connected into implant **162** at recess **180**. This connection may be threaded, magnetic, friction, hex, ball and socket, linkage,

adhesive, and other connections suitable for transferring vibratory energy as disclosed herein or known in the art. Also, other vibratory energy devices as disclosed herein or known in the art may be utilized.

(109) FIG. 30 also shows additional methods of stabilizing a loose implant and/or facilitating the solidification and/or polymerization of bondable material **164**. For example, fastener **170H** may be a metal and/or polymer fastener, which may be affixed to the bondable material **164** and/or implant **162** to stabilize implant **162**. In another example, fastener **170I** may be metal coated with bondable material. Upon the application of vibratory energy and/or heat, distal end **182A** deforms thereby stabilizing the gap between implant **162** and body tissue **160**. In an additional example, fastener **170J** may be made of bondable material. Upon the application of vibratory energy and/or heat, distal end **182B** deforms, thereby stabilizing the gap between implant **162** and body tissue **160**.

(110) Referring to FIG. 31, end effector **104** of instrument **100** may be connected into implant **162** with coupler **184** to stabilize previously hardened and/or polymerized bondable material **164** or to facilitate solidification and/or polymerization of bondable material **164**. This connection may be threaded, magnetic, friction, hex, ball and socket, linkage, adhesive, and other connections suitable for transferring vibratory energy as disclosed herein or known in the art. Also, other vibratory energy devices disclosed herein or known in the art may be utilized.

(111) Referring to FIG. 32, attachment **186** may be attached and/or bonded to implant **162**.

Attachment **186** may be made from any material described herein (i.e. collagen, graft, or growth promoter) or any other material known in the art, preferably to promote healing and/or contain bondable material **164**. For example, vibratory energy may be used to bond attachment **186** to implant **162**. In another example, fastener **170** may secure attachment **186** to implant **162**. In an additional example, vibratory energy may be used to bond fastener **170** to attachment **186**.

(112) Referring to FIGS. 33-34, implant **162** may be manufactured with reservoir **192** or reservoir **192** may be formed during or after implantation. Additionally, therapeutic substance **194** may be incorporated in reservoir **192** of implant **162**, impregnated in implant **162**, or coated on or in implant **162**. As shown in FIG. 33, reservoir **192** may be located in implant **162**. Alternatively, reservoir **194** may be formed in body tissue **160**, as shown in FIG. 34. Cap **188** may be made of bondable material. Additionally, cap **188** may be attached and/or bonded to enclose reservoir **192**. Implant **162**, cap **188**, and/or body tissue **160** may contain attachment feature **190** and/or attachment recess **196** to facilitate mechanical attachment and/or bonding with end effector **104**. Additionally, implant **162**, cap **188**, and/or bondable material **164** may be porous to facilitate the delivery of therapeutic substance **194**.

(113) Referring to FIG. 35, therapeutic substance **194** may be contained in implant **198**, for example drugs or antibiotics contained in an acetabular cup. Implant **198** may be manufactured with reservoir **192A** and/or reservoir **192B** or the reservoirs may be formed during implantation. Any combination of one or more reservoir **192A** and/or reservoir **192** may be used. Cap **188** may be coated with bondable material. Additionally, any of caps **188A-D** may be attached and/or bonded to enclose reservoir **192A** or **192B**, which may provide the potential benefit of multiple release times for therapeutic substance **194**. Caps **188A-D** may contain attachment features **190A-D** to facilitate mechanical attachment and/or bonding. Additionally, implant **198** and/or any of caps **188A-D** may be porous to facilitate the delivery of therapeutic substance **194**.

(114) Referring to FIG. 36, fastener **170** may be used to stabilize implant **162**, for example a tibial component of a total knee arthroplasty (TKA). In an embodiment, fastener **170K** may be bonded to bondable material **164** on the underside of the implant or bonded directly to implant **164**. In another embodiment, fastener **170L** may be bonded to the portion of implant **162** that is within body tissue **160**. In an additional embodiment, fastener **170** may be used to secure tissue graft **202** to implant **162** and/or body tissue **160**. For example, tissue graft **202** may be an allograft. Any embodiment of fastener **170** that has been described herein or known in the art may be used.

(115) Referring to FIGS. 37-38, bondable material **164** may be utilized to stabilize body tissue **160**.

For example, bone cement in previously performed kyphoplasty may become loose and require additional stabilization. In this example, the fasteners may utilize the previously implanted bone cement to stabilize the spine instead of removing and reapplying bone cement. In an embodiment, body tissue **160** has been previously implanted with bondable material **164**. Fastener **170N** is passed through support **168** and bonded to bondable material **164**. As shown in FIG. **37**, one or more fastener **170** is passed through support **168** and secured and/or bonded to surrounding tissue **200**. Fasteners **170M**, **170N**, and **170P** may be any embodiment disclosed herein or known in the art. Additionally, fasteners **170M**, **170N**, and **170P** may be used with embedding fastener **110** as described above. Also in these embodiments, vibratory energy may be used to stabilize previously hardened and/or polymerized bondable material or to facilitate the solidification and/or polymerization of bondable material.

(116) Referring to FIGS. **39-41** and **50-51**, an additional embodiment of instrument **100** includes guide sheath **102**, spring **204**, and/or force regulator **206**. In an embodiment in FIG. **41**, guide sheath **102** may align washer **146** when the tip of end effector **104** is placed in contact with fastener **136**. As fastener **138** is staked and the tip of fastener **136** is shaped with the application of vibratory energy, such as ultrasonic energy, guide sheath **102** may allow end effector **104** to advance while applying force to washer **146**, support **168**, body tissue **160**, and/or bondable material **164** (not shown in FIG. **41**). In a further embodiment in FIG. **41**, guide sheath **102** may hold, guide, align, and/or deliver washer **146**, fastener **136**, or other fasteners referenced herein or known in the art. In an additional embodiment, regulating sheath **206** may have a spring **204**, for example any spring, cushion, or other material or device known in the art for spring/damping applications. Additionally, instrument **100** may have regulating tab **206** for manually applying and/or regulating the movement of guide sheath **102**. Although the embodiment in FIG. **41** may be used under a limitless number of configurations and settings, Table 2 is being set forth with operative examples:

(117) TABLE-US-00002 TABLE 2 Polycarbonate Fastener Bonding Instrument: Handpiece P05 with tuning of 39,000-45,000 Hz System Settings: 39,500 Hz, SOW, 1.0 sec weld time Test Energy Force Deformation Sample Power Application Applied to Depth Number (watts) Time (sec) Break (lbs.) (inches)

1	30	1.69	93.2	0.115	2	28	1.68	86.4	0.110	3	31	1.78	98.2	0.111	4	27	1.80	91.3	0.108
5	31	1.69	109.2	0.109															

(118) In additional embodiments, frequency may preferably be between 20 to 80 khz, power may preferably be between 5 to 200 watts, and energy application time may be preferably between from 0.1 to 5 seconds.

(119) In an embodiment, a sensor may be included in instrument **100**. For example, a force, pressure, or temperature sensor may be used to measure bonding and/or staking. In another example, a visual and/or audio indicator may be operatively connected to the sensor, which may be used to indicate a proper bond/stake. In another embodiment, a visual and/or audio indicator may be connected to instrument **100** or the energy generator, which may be used to illustrate and/or teach proper technique during bonding and/or surgery. In another example, the visual and/or audio indicator may indicate completion of a proper bond/stake, over/under application of force, or expiration of desired energy application time.

(120) In another embodiment, a vacuum may be operatively connected to **100**. For example, the vacuum may be communicatively connected between the guide sheath **102** and end effector **104**, which may be used for the removal of debris from instrument **100**.

(121) Referring to FIGS. **42-43**, an additional embodiment of fastener **170** may include one or more feature **208**. Feature **208** may increase or decrease the transfer of energy across fastener **170**. Feature **208** may pass into the surface or through fastener **170**. Feature **208** may be on any surface or surfaces of fastener **170** and/or contain therapeutic substances. Fastener **170** may include effector interface **210**, preferably for engagement with end effector **104**.

(122) Referring to FIGS. **44-45**, embedding fastener **110** may also include thread **212**. Embedding fastener **110** may be screwed and/or engaged into bondable material, tissue, and/or any other

material disclosed herein, preferably by engaging interface **214** with a screw driver or other tool. Embedding fastener **110** may include an interface **214** which may be radiused, chamfered, funnel-shaped, threaded, or any other shape, for example square, rectangular, circular, elliptical, triangular, hexagonal, or asymmetrical shape. Embedding fastener **110** may be made of any metal, polymer, or other material disclosed herein.

(123) Referring to FIGS. **46-47**, fastener **170** may include effector interface **216**. Preferably for engagement with end effector **104**, effector interface **216** may be radiused, chamfered, funnel-shaped, threaded, or any other shape, for example square, rectangular, circular, elliptical, triangular, hexagonal, or asymmetrical shape.

(124) Referring to FIGS. **48-49**, instrument **100** may include end effector **104** dimensioned and configured for a point of maximum displacement at or near the middle of fastener **170** at or near point **218B** (FIG. **49**). In another embodiment, it may be preferable for instrument **100** to include end effector **104** dimensioned and configured for a point of maximum displacement at or near the end of fastener **170** at or near point **218A** (FIG. **48**). For most applications, a point of maximum displacement at or near point **218B** is preferred. Fastener **170** may be threaded or have an interference fit with end effector **104**.

(125) A point of maximum displacement along end effector **104** may occur at increments of about half its wavelength, which may be determined by the ratio of the speed of sound through the material of end effector **104** to the frequency of the wave propagated through end effector **104**. The end effector **104** may be made of titanium or any material disclosed herein. For example, at a frequency of 20 khz, points of maximum displacement along end effector **104** made of titanium may be in increments of about 4 to 6 inches, preferably 4.8 to 5.1 inches. For example, at a frequency of 40 khz, points of maximum displacement along end effector **104** made of titanium may be in increments of about 2 to 3 inches, preferably 2.4 to 2.5 inches

(126) A point of maximum displacement along fastener **170** may also occur at increments of about half its wavelength. Fastener **170** may be made of PEEK, PLLA, or any material disclosed herein. As an example for PEEK, at a frequency of 20 khz, points of maximum displacement along end effector **104** may be in increments of about 1 to 2 inches, preferably 1.6 to 1.7 inches. As another example for PEEK, at a frequency of 40 khz, points of maximum displacement along end effector **104** may be in increments of about 0.5 to 1 inch, preferably 0.8 inches. As an example for PLLA, as an example at a frequency of 20 khz, points of maximum displacement along end effector **104** may be in increments of about 1 to 2 inches, preferably 1.3 to 1.4 inches. As an example for PLLA, as an example at a frequency of 40 khz, points of maximum displacement along end effector **104** may be in increments of about 0.5 to 1 inch, preferably 0.7 inches.

(127) To optimize bonding and/or reduce the stress applied, the desired point of bonding on fastener **170** should be at or near a point of maximum displacement. For example, the desired point of bonding on fastener **170** may be along half its length or at its tip (see **218A** of FIG. **48**). If the desired point of bonding is at a point of minimal or zero displacement (see **218B** of FIG. **49**), bonding may be difficult. To facilitate bonding, it may be preferable to increase power or amplitude of the signal, thereby increasing energy applied to fastener **170**.

(128) Referring to FIGS. **52-53**, fastener **170T** may be used with sleeve **171T**, potentially to contain and/or release therapeutic substances into a body. Fastener **170T** may engage with sleeve **171T** by mechanical interlock, thread, or vibratory energy bond. End effector **104** may engage with fastener **170T** for vibratory energy bonding. Any fastener **170** disclosed herein may have energy director **222** to facilitate bonding by directing energy to the desired location of bonding.

(129) Referring to FIGS. **54-55**, fastener **170U** may have energy director **224**. Energy director **224** may facilitate bonding with implant **162**, especially if a portion or the entirety of implant **162** includes a porous material. For example, implant **162** may include a porous metal. End effector **104** may engage with fastener **170U** for vibratory energy bonding.

(130) Referring to FIG. **56**, fastener **170F** may be embedded into a bondable material with its

leading end and stabilize a support **168** (i.e. plate) with its trailing end. End effector **104** may engage with fastener **170F**, preferably near the trailing end, for vibratory energy bonding. Fastener **170F** may include any material disclosed herein, but preferably titanium or titanium with at least a portion coated with PEEK or PLLA.

(131) Referring to FIGS. **57-58**, fastener **170W** may be used with, preferably disposed within, expanding anchor **226**. End effector **104** may engage with fastener **170W**, preferably near the trailing end, for vibratory energy bonding. A hole in tissue and/or bondable material may be formed or drilled into body tissue **160** prior to or during the implantation of fastener **170W** and expanding anchor **226**. Fastener **170W** and expanding anchor **226** may pass into body tissue **160**, for example in the configuration shown in FIG. **57**. Fastener **170W** may be retracted into expanding anchor **226**, preferably after being positioned in body tissue **160**. Expanding anchor **226** may expand outwards (shown in FIG. **58**), thereby engaging and/or exerting a radially outward force on body tissue **160**. Preferably after fastener **170W** and expanding anchor **226** are in an expanded condition, vibratory energy may be applied to fastener **170W**, preferably near the trailing end, to bond fastener **170W** and expanding anchor **226** together. In another embodiment, fastener **170W** and/or expanding anchor **226** may be configured to bond into a bondable material. After fastener **170W** and expanding anchor **226** have been stabilized in the expanded configuration, the excess length of the trailing end of fastener **170W** may be removed to be substantially flush with the trailing end of expanding anchor **226**.

(132) Referring to FIG. **59**, fastener **170** may be used to increase interference between body tissue **160** and implant **198**. Fastener **170** may be an interference screw and/or used in conjunction with interference implant **228** to position and/or stabilize implant **198**. Fastener **170** may pass through all or a portion of implant **198**. In additional embodiments, fastener **170** may stabilize body tissue **160** (i.e. ACL graft) against the side of a hole in body tissue **160** (i.e. bone) (not shown), stabilize body tissue **160** (i.e. soft tissue) to another body tissue **160** (i.e. bone), or stabilize interference implant **228** (or tissue graft **202**) to body tissue **160** (i.e. bone) and/or implant **198** (FIG. **59**). In another embodiment, fastener **170** may be part or entirely made of a biodegradable and/or bondable material. In an embodiment, fastener **170** may have a snap that would overlay part of implant **198** for interference. In another embodiment, implant **198** may include porous surface **230** (FIG. **59**) or a coating of bondable material. In another example, implant **198** may be bonded with vibratory energy, hydrophilic, and/or mechanically expandable against body tissue **160** (i.e. bone), which may allow the implant to sequentially expand and provide interference against body tissue **160** (i.e. bone) or another implant.

(133) In an embodiment related to hip (or shoulder) resurfacing, implant **198** may be an acetabular component or cup (or glenoid component), which is commonly stabilized using screws through its center. To replace the use of these screws or to provide additional stabilization, implant **198** may be stabilized by positioning fastener **170** between implant **198** and body tissue **160** (i.e. acetabulum or glenoid), which may urge implant **198** to the desired position and/or enhance interference with body tissue **160**. In an embodiment, implant **198** may be free of holes, as fastener **170** may provide the majority of interference.

(134) There are many different features to the present invention and it is contemplated that these features may be used together or separately. Thus, the invention should not be limited to any particular combination of features or to a particular application of the invention. Further, it should be understood that variations and modifications within the spirit and scope of the invention may occur to those skilled in the art to which the invention pertains. Accordingly, all expedient modifications readily attainable by one versed in the art from the disclosure set forth herein that are within the scope and spirit of the present invention are to be included as further embodiments of the present invention.

Claims

1. A vibratory energy system for bonding a first component to a second component, comprising: a hand-held instrument comprising: a proximal end configured to couple to an energy source to enable the hand-held instrument to receive energy from the energy source; a vibratory energy generator configured to generate mechanical vibratory energy from the energy received from the energy source and to control application of the mechanical vibratory energy; a force regulating mechanism configured to control application of force applied by the vibratory energy generator; a sensor configured to measure at least one of force or pressure applied by the vibratory energy generator during the bonding of the first component to the second component; a housing enclosing the vibratory energy generator, the force regulating mechanism, and the sensor, the housing having an external gripping surface for a human hand; and a switch coupled to the housing and operatively connected to the vibratory energy generator, the switch configured to activate the vibratory energy generator; and an end effector configured to attach to a distal end of the hand-held instrument and operatively connected to the vibratory energy generator, wherein the end effector is configured to contact the first component; and an indicator operatively connected with the sensor and configured to provide one or more of a visual indication of the bonding or an audible indication of the bonding based on the at least one of force or pressure measured by the sensor during the bonding of the first component to the second component, wherein during contact of the first component with the second component, the hand-held instrument is operable to apply the mechanical vibratory energy from the vibratory energy generator through the end effector to the first component to bond the first component to the second component.
2. The vibratory energy system of claim 1, wherein the first component is formed, at least in part, from a first polymer and the second component is formed, at least in part, from a second polymer that is chemically the same as the first polymer.
3. The vibratory energy system of claim 1, wherein the first component is formed, at least in part, from a first polymer and the second component is formed, at least in part, from a second polymer that is chemically different from the first polymer.
4. The vibratory energy system of claim 1, wherein the first component is formed, at least in part, from a polymer and the second component is formed, at least in part, from a metal.
5. The vibratory energy system of claim 1, wherein the mechanical vibratory energy is ultrasonic energy, and wherein the force regulating mechanism comprises a dampening component.
6. The vibratory energy system of claim 1, wherein the end effector is formed, at least in part, from titanium.
7. The vibratory energy system of claim 1, wherein the end effector comprises a connector to couple to the first component prior to bonding and to decouple the end effector from the first component after the bonding.
8. The vibratory energy system of claim 1, wherein the end effector is configured to be detachable from the distal end of the hand-held instrument, and wherein the hand-held instrument is configured to employ the end effector dependent on material properties of at least one of the first and second components.
9. The vibratory energy system of claim 1, further comprising a guide sheath configured to align with the end effector during the bonding of the first component to the second component and to apply a force to the first component, wherein the guide sheath is further configured to align the end effector at a desired point of bonding of the first component to the second component, and wherein the hand-held instrument comprises a manual regulating tab for manually regulating movement of the guide sheath.
10. A vibratory energy system for bonding a first component to a second component, comprising: an energy source; a user interface coupled to the energy source; and a hand-held instrument

comprising: a proximal end configured to couple to the energy source to enable the hand-held instrument to receive energy from the energy source; a vibratory energy generator configured to generate mechanical vibratory energy from the energy received from the energy source and to control application of the mechanical vibratory energy; a force regulating mechanism configured to control force applied by the hand-held instrument; a sensor configured to measure at least one of force or pressure applied by the hand-held instrument during the bonding of the first component to the second component; a housing enclosing the vibratory energy generator, the force regulating mechanism, and the sensor, the housing having an external gripping surface for a human hand; and a switch coupled to the housing and operatively connected to the vibratory energy generator, the switch configured to activate the vibratory energy generator; and an end effector configured to attach to a distal end of the hand-held instrument and operatively connected to the vibratory energy generator, wherein the end effector is configured to contact the first component; an indicator operatively connected with the sensor and configured to provide one or more of a visual indication of the bonding or an audible indication of the bonding based on the at least one of force or pressure measured by the sensor during the bonding of the first component to the second component, wherein during contact of the first component with the second component, the hand-held instrument is operable to apply the mechanical vibratory energy from the vibratory energy generator through the end effector to the first component to bond the first component to the second component.

11. The vibratory energy system of claim 10, wherein the mechanical vibratory energy is ultrasonic energy.

12. The vibratory energy system of claim 10, wherein the force regulating mechanism comprises a dampening component, and wherein the end effector comprises a connector to couple to the first component prior to bonding and to decouple the end effector from the first component after the bonding.

13. The vibratory energy system of claim 10, wherein the end effector is detachable, and wherein the hand-held instrument is configured to the end effector dependent on material properties of at least one of the first and second components.

14. The vibratory energy system of claim 10, wherein the end effector is detachable, and wherein the hand-held instrument is configured to employ the end effector dependent on configurations of at least one of the first and second components.

15. The vibratory energy system of claim 10, further comprising a guide sheath configured to align the end effector centered on a desired point of bonding of the first component to the second component.

16. A vibratory energy system for bonding two or more materials, comprising: an energy source; a hand-held vibratory energy application device comprising a housing having an external surface configured to be held by a human hand, the vibratory energy application device coupled to the energy source and comprising a vibratory energy generator configured to generate mechanical vibratory energy, wherein the vibratory energy generator is enclosed within the housing; a switch coupled to the hand-held vibratory energy application device and configured to control activation of the mechanical vibratory energy for the bonding of the two or more materials; a second control mechanism incorporated into the hand-held vibratory energy application device and configured to control a force transmitted by the hand-held vibratory energy application device; and at least one of a force sensor or a pressure sensor configured to output one or more signals indicative of at least one of the force or the pressure applied by the hand-held vibratory application device during the bonding between the two or more materials; and an indicator operatively connected with the sensor and configured to provide one or more of a visual indication of the bonding or an audible indication of the bonding based on the at least one of force or pressure measured by the sensor during the bonding between the two or more materials.

17. The vibratory energy system of claim 16, comprising a regulatory mechanism incorporated into

the vibratory energy application device and configured to regulate force applied through the vibratory energy application device to the two or more materials.

18. The vibratory energy system of claim 16, wherein the hand-held vibratory energy application device is configured for at least one detachable end effector, wherein the detachable end effector is configured to provide bonding according to configurations of the two or more materials.

19. The vibratory energy system of claim 16, wherein the hand-held vibratory energy application device is configured for at least one detachable end effector, wherein the detachable end effector is configured to provide bonding according to chemical properties of the two or more materials.

20. The vibratory energy system of claim 16, further comprising: a coupler configured to fix the first material relative to the second material, and wherein the second control mechanism comprises a damping component incorporated into the hand-held vibratory application device.
